Manual

FINA API

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TOrmMethod

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Security

- Basic Authentication
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Performance

- I. Compress
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- III. JSON Compression (JSON Object ke Array of Records)
- IV. JSON Decompression (Array of Records ke JSON Object)

History

Notes

- **Minimum FINA API 1.0.4.21-14908**
- **For TJSONValue input:**
  - For ”id” is required for unmarshall.
  - The field with the prefix “F” can be modified.
  - For TDateTime Compatibility, use can use https://stackoverflow.com/questions/57184472/javascript-date-to-delphi-tdatetime#:~:text=A%20JavaScript%20Date%20object%20is,to%20produce%20a%20TDateTime%20value.

    ```
    function jsDateToDelphiDate(dateToConvert){
      const UnixDateDelta = 25569.0;
      const SecsPerDay = 86400;
      const MSecsPerSec = 1000;
      var UnixSeconds = dateToConvert.getTime() / MSecsPerSec; // 1563984000
      var SecsToAdd = UnixSeconds / SecsPerDay; // 18101.666666666668
      return UnixDateDelta + SecsToAdd;
    }
    ```

REST URL

GET

http://[IP]:212/fina/rest/[class-name]/[method-name]

Example:
IP = 192.168.1.100
Class-Name = TMultiDBServerMethod
Method-name = orm
The URL should be: **http://192.168.1.100:212/fina/rest/TOrmMethod/orm/param1/param2**

POST

For the POST method, the method name should be enclosed with %22.
For example: **http://192.168.1.100:212/fina/rest/TOrmMethod/%22orm%22**

All of the parameters should be sent in the body in the format of:
{_parameters:[“param1”, “param2”,...]}
If the parameter is a JSON, it should be stringified.

If there is only one parameter, it can be sent directly in the body.

TMultiDBServerMethod

function ExecuteSQL

1. Purpose: To select an SQL query into one or more databases.
2. Parameters:
    a. const sql:String → the SQL to be executed
    b. Dataset: String → the database index to be query upon. If more than one, separate using comma (e.q: ‘1,2’)
3. Return : string → queried data in json format

function ExecuteSQLWithParameters

1. Purpose: To execute an SQL using parameter
2. Parameters:
    a. paramJSON : string
3. Return : string

function loadReportList

1. Purpose: To load IT inventory report list from ini file
2. Parameters: none
3. Return : TJSONObject

function getReportData

1. Purpose:
2. Parameters:
    a. dbName:string
    b. selectedReport : string
3. Return : TJSONObject

TFinaMethod

function ImportJson

1. Purpose: To import from json string to FINA Datamodule
2. Parameter
    a. const ToDatabase: Integer → database index
    b. const UserID: Integer → FINA userID
    c. const JsonString:String → json object of the data to be imported into FINA
3. Return: String → if success: empty string, otherwise get an error message from FINA.

Note: ImportJson depends on MapField.ini.

function json2fina

1. Purpose: To insert or update data to datamodule in FINA
2. Parameters:
    a. json : string
3. Return: TJSONObject
    a. If success: {result:[{id: transactionid}]}
    b. If failed: {result:[{error: [Error Message]}]}

Format data

```
{"_parameters":["{
      \"id\"          : number|string,
      \"userId\"      : number,
      \"dataModule\"  : string,
      \"transType\"   : string,
      \"dbIndex\"     : number,
      \"delete\"      : 1 | 0 (default = 0),
      \"data\":{
            \"master\":{
                  \"field1\":xxx,
                  \"field2\":yyy,
                  \"extended.customfield1\":zzz
                  ...,
                  \"detail\":[
                        {
                              \"field_a\":xxx,
                              \"field_b\":yyy,
                              ...
```

```
                        },
                        ...
                        {
                              “seq”:xx,
                              “delete”:1
                        },
                        {
                              \”field_a\”:xxx,
                              \”field_b\”:yyy,
                              ...
                              "detail": [{
                                                    \"discaccount\":
      Number,
                                                \"discount\": Number
                                          }]
                        }
                  ]
            },
      }
}”]}
```

Example: payload for CR Transaction:


```
{
   "_parameters": [
      {
         "id": -1,
         "dataModule": "TdmARPayment",
         "transType": "ARPMT",
         "userId": 0,
         "dbIndex": 3,
         "data": {
            "master": {
               "billtoid": 8660,
               "bankaccount": "112.01",
               "chequedate": "2024-06-28",
               "chequeamount": 11720448,
               "paymentdate": "2024-06-28",
               "chequeno": "cek1001",
               "cheque": 1,
               "description": "Coba Input Cust Recv dari Biz-A",
               "detail": [
                  {
                     "arinvoiceid": 110173,
                     "paymentamount": 9246752,
```

```
                "disctakenamount": 10000,
                "detail": [
                  {
                    "discaccount": "122.01",
                    "discount": 10000
                  }
                ]
              },
              {
                "arinvoiceid": 110175,
                "paymentamount": 2473696,
                "disctakenamount": 20000,
                "detail": [
                  {
                    "discaccount": "122.01",
                    "discount": 20000
                  }
                ]
              }
            ]
          }
        }
      }
    ]
}
```


Explanation

1. Because the json data is inside the parameters value, so every double quote must be escaped (\").
2. Key name is case sensitive except for the field names.
3. **dbIndex** is a parameter to determine which database to be modified.
4. **userId** FINA's credential.
5. **id**:
    a. -1 = insert. If the primary key of the table is string, use '' (empty string).
    b. Other than -1 means edit mode.
6. To delete a record, set key **delete :1**. If **delete** = 0, the transaction will do nothing.
7. For all fields, refer to each table's field name.
8. For extended fields, start with **extended.** (i.e.: extended.customfield1)
9. For editing detail:
    a. make sure the first key is the unique key
    b. If unique key does not exist in **detail** object, no changes will be made
10. To delete a detail, provide an object consist of:
    a. Unique key
    b. delete: 1 key.

11. When updating a MD record, first all old details will be deleted. After that, all new details will be inserted. No edit will be performed.

List of Available dataModules

Transaction Name	DataModule Name	transType
Journal Voucher	TdmJV	GLJV
Other Payment	TdmJV	GLOP
Other Deposit	TdmJV	GLOD
Sales Order	TdmSalesOrder	ACSOR
Delivery Order	TdmARInvoice	ARDLV
Sales Invoice	TdmARInvoice	ARINV
Sales Invoice from DO	TdmARInvoice	ARSTM
Customer Receipt	TdmARPayment	ARPMT
Sales Return	TdmCreditMemo	ARRTN
Sales Return from DO/SI	TdmCreditMemo	ARRTR
Purchase Order	TdmPurchaseOrder	APPO
Receive Item	TdmAPInvoice	APRCV
Purchase Invoice	TdmAPInvoice	APINV
Vendor Payment	TdmAPCheque	APCHQ
Purchase Return	TdmDebitMemo	APRTN
PR from RI	TdmDebitMemo	APRTC
PR from PI	TdmDebitMemo	APRTR
Item Transfer	TdmStockTrans	ITTFR
Item Request	TdmItemRequest	ITREQ
Job costing	TdmJobCost	PRJOB
Job Costing Roll Over	TdmJCRO	JCRO
Inventory Adjustment	TdmAdjustItem	ITADJ

Selling Price Adjustment	TdmItemPriceAdj	ITSPA
Item	TdmItem	ACITM
Customer	TdmCustomer	ACCUS
Vendor	TdmVendor	ACVDR
Project	TProjectDataset	ACPRJ
Department	TDepartmentDataset	ACDEP
Fixed Asset	TdmFixedAsset	FAACQ
Fiscal Fixed Asset Type	TdmFisFAType	FAFIS
Fixed Asset Type	TdmFixAssetType	FATYP
GL Account	TdmGLAccount	ACACC
Item Grouping	TdmAssembly	ITGRP
Currency	TCurrencyDataset	
Customer Type	TCustTypeDataset	
Item Category	TItemCategoryDataset	
Salesman	TSalesmanDataset	
Shipment	TShipmentDataset	
Tax	TTaxDataset	
Template	TTemplateDataset TTemplateDetDataset	
Term of Payment	TTermDataset	
User	TUserApiDataset	
Warehouse	TWarehouseDataset	
Default Account	TDefaultAccountDataset	
Extended Type	TExtendedTypeDataset	
Extended Mapping	TExtendedMappingDataset	
Extended Detail	TExtendedDetDataset	
Extended	TExtendedDataset	

Project Budget	TProjectBudgetDataset	
Department Budget	TDepartmentBudgetDataset	
Opening Balance Dept/Proj	TDeptProjectOBDataset	
Commission	TCommissionDataset	
Contact	TContactDataset	
Item Commission	TItemCommissionDataset	
Salesman Commission	TSalesmanComissionDataset	
Customer By Salesman	TCustBySalesDataset	
Salesman By Item Category	TSalesByICatDataset	

Example

Json data for Journal Voucher:


```
{"_parameters":["{
	\"publicIp\":\"localhost\",
	\"id\":\"78\",
	\"dataModule\":\"TdmJV\",
	\"transType\":\"GLJV\",
	\"dbIndex\":\"1\",
	\"data\":{
		\"master\":{
			\"JVNumber\":\"78\",
			\"TransDate\":\"2020-06-11\",
			\"detail\":[
				{
					\"GLAccount\":\"10000\",
					\"GLAmount\":\"5500\",
				},
				{
					\"GLAccount\":\"50000\",
					\"GLAmount\":\"-5500\",
				}
			]
		}
	}
}"]}
```

JSON for TdmJCRO to Item

```
{
  "id":-1,
  "dataModule":"TdmJCRO",
  "transType":"",
  "userId":"0",
  "dbIndex":1,
  "data":{
    "master":{
      "jcid": [mfsht.mfshtid],
      "rotype":[1=to account | 2=to item],
      "json":"{
        \"id\":-1,
        \"dataModule\":\"TdmAdjustItem\",
        \"transType\":\"ITADJ\",
        \"dbIndex\":1,
        \"userId\":0,
        \"data\":{
          \"master\":{
            \"adjno\":\"XXXXX\",
            \"adjaccount\":\"XXXXX\",
            \"adjDate\":\"YYYY-MM-DD\",
            \"adjcheck\":1,
            \"description\":\"test\",
            \"warehouseid\":0,
            \"mfid\":[mfsht.mfshtid],
            \"detail\":[
              {
                 \"itemno\":\"xxxxxx\",
                 \"qtydifference\":1,
                 \"valuedifference\":1000,
                 \"deptid\":202,
                 \"allocated\":100,
                 \"itemreserved1\":\"SN-004\"
              }
            ]
          }
        }
      }"
    }
  }
}
```

JSON for TUserApiDataset

To Create new user:

```
{
  "id":-1,
  "dataModule":"TUserApiDataset",
  "transType":"",
  "userId":"0",
  "password":"[password_of_userId]",
  "dbIndex":1,
  "data":{
    "master":{
      "username": "",
      "userlevel":[0=administrator | 1=supervisor | 2=user],
      "fullname": "",
      "plainpassword": "[provide unencrypted password]",
      .. .. [Fields in USERS TABLE],
    }
  }
}
```

POST template

- Purpose: to get a rendered template
- Body:

```
{
  "dbIndex"           :number,
  "id"                :number,
  "table"             :string,
  "keyfield"          :string,
  "templateId"        :number,
  "userId"            :string (default = -1),
  "imageQuality"           :number (default = 90),
  "dictionary"  :[
     {
       "param":string,
       "value":string
     },
     …
  ]
}
```

- Return: stream (array of bytes) of jpeg image. Convert it to base64 image. If there is more than 1 image, it must be parsed based on the initial image marker.

Explanation:

1. **dbIndex** is the database index to be used.
2. **id** is the id of the form to be previewed.
3. **table** is the table name of the form to be previewed.
4. **keyfield** is the field name that hold the primary key of the table.
5. **templateId** can be filled with template id from table TEMPLATE in FINA DB.
6. **userId (optional,** `default -1 mean not active user`**)** is the id of FINA user to be uses at report function
7. **imageQuality (optional,** `default 90`**)** range from 1-100. Higher quality results in a better picture, but a bigger data size
8. **dictionary** is the array of objects to be assigned in each form. The object must consist of these keys:
    a. param: the key to be set in dictionary
    b. value: is the value to be set

List of dictionary to be set in each Form:

Form Name	Dictionary
SO	SOID MaxInvDet

POST calcDiscount

- Purpose: To calculate discount promo
- Body: Same as json2fina for TdmSalesOrder & TdmARInvoice
- Return:

```
{
  detail:[
    {
      itemno:string,
      itemovdesc:string,
      quantity:number,
      unitprice:number,
      discount:string
    },
    …
  ]
}
```

Explanation:

1. calcDiscount can be used for Sales Order and Sales Invoice transactions only.
2. For body format, see Function json2fina.
3. Mandatory field in master:
    a. Customer id
    b. Date
    c. Term of payment
4. Mandatory field in detail:
    a. Item number
    b. Quantity
    c. Unit price
    d. Item unit
    e. Item discount
5. Mandatory field in detail (for Distro):
    a. Extended.CustomField1 to 6

POST fml2fina

- Purpose: To import fml file to fina
- Body:

```
{
  dbIndex : number,
  userId  : number,
```

```
  showError : number | boolean,
  fml     : string
}
```

- Return:

```
{
  success : true|false
}
```

- FML Examples:

```
<AUDITXML TOTALTRANSACTION = "1">
    <PROJECTBUDGET Status="Edit"
ID="{25E211B3-B282-4F37-9C69-F246B82200E9}"
TransactionType="PRBGT">
        <Period1 OldValue="456" NewValue="999" />
        <USERGUID Value="{2C9F81A5-F566-470A-AED4-C6ABB34739C6}"
/>
        <FML_GUID Value="{C19499D3-2ABB-4652-A521-2769EB7A5989}"
/>
    </PROJECTBUDGET>
</AUDITXML>

or

<AUDITXML TOTALTRANSACTION = "1">
     <JV ID = "{2057664E-950C-42D4-871C-A17820621A79}"
TransactionType = "GLJVO" Status = "Delete">
          <USERGUID
     Value="{2589F978-0B19-4890-89D5-81161861034B}" />
          <FML_GUID
     Value="{C19499D3-2ABB-4652-A521-2769EB7A5989}" />
     </JV>
</AUDITXML>
```

- Info:

<AUDITXML> is needed to wrap Transaction / Master data.
AUDITXML has attribute TOTALTRANSACTION.
USERGUID is user GUID. The data can be gathered from USERS, field GUID.
FML_GUID is GUID generated (can be generated with crypto.randomUUID().toUpperCase()). To set a unique GUID in AUDITXML field GUID.

Explanation:

1. To know what is the format of FML (Fina Markup Language), try to insert any transaction in FINA and export the FML file.
2. This function will import FML files as if we import FML files directly from FINA.

TMfgMethod

To use this function, make sure you have set multiserver setting for
Main | Database_FINA

function SaveRMOut

- Purpose:
- Parameters:
    - const RMOuts: TRawMaterialOuts
    - Tx:TjbTx=nil
- Return:String

function SaveFGIn

- Purpose:
- Parameters:
    - const FGIn: TFGIn
    - Tx:TjbTx=nil
- Return:Integer

function SaveCloseWO

- Purpose:
- Parameters:
    - const CloseWO:TCloseWO
    - Tx:TjbTx=nil
- Return:Boolean;

function LoadOutstandingSchedule

- Purpose:
- Parameters:
    - const Tx:TjbTx
- Return:TStringList (Scheduleid, Fgcode, Fgqty, Scheduledate, Shift, Itemno, Itemunit, Quantity, Fgdesc, Rmdesc, Fgitemunit, Psitemunit, Psqty, Rmqty)

function SaveStartProduction

- Purpose:
- Parameters:
    - const ASchedule:TSchedule

    - LotNos:String
    - Tx:TjbTx
- Return:String

function SaveRMAdj

- Purpose:
- Parameters:
    - const RMAdjs : TRawMaterialAdjs
    - Tx:TjbTx
- Return:String

Function SaveRMOutMarshal

- Purpose:
- Parameters:
    - RMOuts:TJsonValue
      Example:

```
{
  "_parameters": [
    {
      "type": "RawMaterialOut.TRawMaterialOuts",
      "id": 1,
      "fields": {
        "FListHelper": [
          {
            "type": "RawMaterialOut.TRawMaterialOut",
            "id": 2,
            "fields": {
              "FProductionID": 1,
              "FScheduleID": 5,
              "FQtyToTake": 4,
              "FCode": "INV001",
              "FItemUnit": "PCS",
              "Fbatch_number": "",
              "FCreatedDate": 45385.8910099306,
              "FUserID": 0,
              "FWhID": 4 <optional>
            }
          }
        ]
      }
    }
  ]
```

}
Info:
-source: InsertRMOut
-minimalist examples, unused information has been deleted.

- Return:String

function SaveCloseWOMarshal

- Purpose:
- Parameters:
  - data:TJsonValue
    Example:
    {
      "_parameters": [
        {
          "type": "CloseWO.TCloseWO",
          "id": 1,
          "fields": {
            "FWOID": 1101,
            "FCreatedDate": 45075.5015595023,
            "FUserID": 0
          }
        }
      ]
    }
  - Info:
    -source TestCloseWO
    -minimalist examples, unused information has been deleted.
- Return:Boolean

procedure doAutoCloseWO

- Purpose:
- Parameters:
  - ds:TDataset
- Return: none

function SaveFGInMarshal

- Purpose:
- Parameters:
  - Data: TJsonValue
    Example:
    {

```
"_parameters": [
 {
  "type": "FGIn.TFGIn",
  "id": 1,
  "fields": {
   "FScheduleID": 5,
   "FFGQty": 10,
   "FBSQty": 2,
   "FResiduQty": 1,
   "FLotNo": 999,
   "FNotes": "",
   "FCreatedDate": 45441.4904470833,
   "FUserID": 0,
   "FWhID": 4 <optional>
  }
 }
]
}
```


- Info:
  -source TestFGIn
  -minimalist examples, unused information has been deleted.

- Return: Integer

function LoadOutstandingScheduleInJsonFormat

- Purpose:
- Parameters:
- Return:String

function SaveStartProductionMarshal

- Purpose:
- Parameters:
  - Data : TJsonValue
  - LotNos:String, ex:"lot1,lot2,..."
    Example:

```
{
 "_parameters": [
  {
   "type": "Schedule.TSchedule",
   "id": 1,
   "fields": {
    "FMachine": {
     "type": "Machine.TMachine",
```

```
      "id": 2,
      "fields": {
        "FID": 11
      }
    },
    "FID": 17183,
    "FUserID": 0
  }
 },
 ""
]
}
```

Info:
-source: TestSaveStartProduction
-minimalist examples, unused information has been deleted.

- Return:String

function LoadWOFromSchedule

- Purpose: Find Schedule Info.
- Parameters:
  - const ScheduleID:Integer
  - Tx:TjbTx=nil
- Return:String

function LoadWOInfo

- Purpose:
- Parameters:
  - const WONo:String
- Return:String;

function SaveRMAdjMarshal

- Purpose
- Use \n to add more than 1 RMAdj in 1 IA.:
- Parameters:
  - const RMAdjList:String
    Format :
    "ScheduleID^RMNo^Quantity^ItemUnit^User_ID^batch_number^reason^"
    Example:

    ```
    {
      "_parameters": [
        "48372^AC7000BM#WIP#^7^PCS^198^JK457065^Patah^"
    ```


```
    ]
}

More than 1 RMAdj in 1IA.
{
  "_parameters": [
    "76^RM D^1^KG^0^^\n76^RM E^2^KG^0^^"
  ]
}
```

- Return:String;

function SearchItem

- Purpose:
- Parameters:
  - const ItemNo: String
  - var ItemDesc:String
  - Unit1:String
  - Unit2:string
  - Unit3: String
- Return: Boolean

function LoadRouteInfo

- Purpose:
- Parameters:
  - const ScheduleID: Integer
  - var FGCode:string
  - BSCode:String
  - ResiduCode: String
- Return: Boolean

function SaveProdResultMarshal

- Purpose:
- Parameters:
  - const CreatedDate:TDateTime (Format :”yyyy-mm-dd**T**hh:mm:ss”)
  - const ID:Integer (Fill with -1 to ID auto generate)
  - ScheduleID:Integer
  - UserID:Integer
  - ProdQty:Currency
  - BSQty:Currency
  - ResiduQty:Currency
  - GrossQty:Currency

- TareQty:Currency
- BSItem :String
- BSReason :String
- ResiduItem:String
- ResiduReason:String
- LotNo:TLotNo (Integer), can be obtained from GetNextLotNo Service.
- Notes : String
- FromLotNo:String, if fromLotNo more than 1, use \n as a separator.
- MachineID:Integer
- Inputted_item_unit : string

  Example:

```
{
  "_parameters": [
     "2024-05-29T00:00:00",
     -1,
     18147,
     8,
     10,
     0,
     0,
     0,
     0,
     "",
     "",
     "",
     "",
     250944,
     "2#73#17# RC",
     "",
     9,
     "KG"
  ]
}
```

- Notes: For BS and Residu, if no BS or Residu, Set BSItem or ResiduItem with empty string “” and set BSQty or ResiduQty with 0.
- Return:Integer with const:

```
msg_unknown_error          = 0;
msg_Success                = 1;
msg_Fail_to_save           = 2;
msg_BSCODE_NotFound        = 3;
msg_RESIDUCODE_NotFound    = 4;
```

function LoadWasteItem

- Purpose:
- Parameters: none
- Return:String

function LoadWasteReason

- Purpose:
- Parameters: none
- Return:String

function CreateWOFromSO

- Purpose:
- Parameters:
    - data:String
      Format :
"SOID^SOSeq^ItemNo^ProductionOrderQty^ItemUnit^RouteID^UserID^IREQID^IREQSeq^WODate^WONO^"
      Example:
      {
        "_parameters": [

      "11343^0^CMS4P-050-0920-GP^15235^KG^36164^9^NULL^NULL^05.09.2023, 08:28:20.776^WO1^"
        ]
      }
- Description:
    - WONO is optional. If the value = ‘’, number will be given based on next running number
- return:String

function SaveSchedule

- Purpose:
- Parameters:
    - WOListData : TJSONValue
      Example:
      {
        "_parameters": [
          {
            "type": "WorkOrder.TWorkOrders",
            "id": 1,

```
    "fields": {
      "FListHelper": [
        {
          "type": "WorkOrder.TWorkOrder",
          "id": 2,
          "fields": {
            "FSchedules": {
              "type": "Schedule.TSchedules",
              "id": 3,
              "fields": {
                "FListHelper": [
                  {
                    "type": "Schedule.TSchedule",
                    "id": 4,
                    "fields": {
                      "FFGQuantity": 10,
                      "FMachine": {
                        "type": "Machine.TMachine",
                        "id": 5,
                        "fields": {
                          "FID": 90
                        }
                      },
                      "FItemUnit": "KG",
                      "FScheduleDate": 45384.5649853472,
                      "FFGCode": "INV001",
                      "FShift": 1,
                      "Fprocess_id": 12,
                      "FstartTime": 45384.5649853472,
                      "FfinishTime": 45384.5649853472
                    }
                  }
                ]
              }
            },
            "FID": 100,
            "FUserID": 0
          }
        }
      ]
    }
  }
]
}
```

Info:
-source: sortWorkCenterByProcessSequence | createWorkOrder.
-minimalist examples, unused information has been deleted.

- return:TJSONArray

function CreateRequestRM

- Purpose:
- Parameters:
  - data:String
    Format:"ScheduleID^RequestDate^Code^QtyToTake^ItemUnit^UserID^"
    Example:
    {
      "_parameters": ["17191^05.09.2023,
    08:25:55.451^CA-UF1810S1@25KG^3600.0305^KG^9^"
      ]
    }
- return:TJSONArray

function ConfirmFGIsQCed

- Purpose:
- Parameters:
  - const ScheduleID:Integer
  - const Qty:Currency
  - const LotNo:TLotNo
- Return: Integer

function LoadMachinesBySchedule

- Purpose:
- Parameters:
  - const ScheduleID:Integer
- Return:String

Function GetScheduleMachineID

- Purpose:
- Parameters:
  - const ScheduleID:Integer
- Return:Integer

function ScheduleIsFirstProcess

- Purpose:
- Parameters:
    - const ScheduleID:Integer
- Return:Boolean

function LoadQCReason

- Purpose:
- Parameters:
- Return:String

function GetNextLotNo

- Purpose: to get next lot no from generator
- Parameters: -
- Return:Integer

function LoadOutstandingLotNo

- Purpose:
- Parameters:
    - const AScheduleID:TScheduleID
    - test_tx:TjbTx=nil
- Return:String

function IsLastProcess

- Purpose:
- Parameters:
    - const AScheduleID: TScheduleID
- Return: Boolean

function SaveQC

- Purpose:
- Parameters:
- ff
    - data:String
      Format:"User_ID^ScheduleID^LotNo^Status^RejectReasonID^note^"
      Example:
      {
        "_parameters": [

```
        "8^18147^250944^1^-1^^"
        ]
    }
```

- Notes:
  Set RejectReasonID to -1 if no reason
  Set Note to empty string if no note
- Return:String

function LoadOutstdLotNoFromPrevProcess

- Purpose:
- Parameters:
  - const AScheduleID:TScheduleID
- Return:String

function LoadQCPassedLotNo

- Purpose:
- Parameters:
  - const AScheduleID:TScheduleID
- Return:String

function LoadOutstandingMachines

- Purpose:
- Parameters:
  - const AScheduleID:TScheduleID
  - test_tx:TjbTx=nil
- Return:String

function LoadOutstandingStartLotNo

- Purpose:
- Parameters:
  - const AScheduleID:TScheduleID
  - test_tx:TjbTx=nil
- Return:String

function LoadQCOutstandingLotNo

- Purpose:
- Parameters:
  - const AScheduleID:TScheduleID
  - test_tx:TjbTx=nil

- Return:String;

function delete_machine

- Purpose:
- Parameters:
    - const machine_id:integer
    - test_tx:TjbTx=nil
- Return:string

function delete_work_center

- Purpose:
- Parameters:
    - const wc_id:integer
    - test_tx:TjbTx=nil
- Return:string

function get_new_primary_key_for_process_table

- Purpose:
- Parameters:
- Return:integer

function get_new_primary_key_for_route_table

- Purpose:
- Parameters:
- Return:integer

Function get_new_primary_key_for_processRawMaterial_table

- Purpose:
- Parameters:
- Return:integer

function do_save_route

- Purpose:
- Parameters:
    - const ARoute:TRoute
- Return:string

function save_route

- Purpose:
- Parameters:
  - const route_data:TJsonValue
  - test_tx:TjbTX=nil
    Example:


```
{
   "type": "Route.TRoute",
   "id": 1,
   "fields": {
      "FFGCode": "INV001",
      "FDescription": "test",
      "FProcesses": {
         "type": "Process.TProcesses",
         "id": 2,
         "fields": {
            "FRoute": {
               "type": "Route.TRoute",
               "ref": 1
            },
            "FOwnsObjects": true,
            "FListHelper": [
               {
                  "type": "Process.TProcess",
                  "id": 3,
                  "fields": {
                     "FRoute": {
                        "type": "Route.TRoute",
                        "ref": 1
                     },
                     "FOutputQuantity": 10,
                     "FWorkCenter": {
                        "type": "WorkCenter.TWorkCenter",
                        "id": 4,
                        "fields": {
                           "FWorkCenterName": "",
                           "FMachines": {
                              "type": "Machine.TMachines",
                              "id": 5,
                              "fields": {
                                 "FOwnsObjects": true,
                                 "FListHelper": []
                              }
```

```
        },
        "FNumberOfShift": 0,
        "Fwork_shift": {
          "type": "WorkCenterShift.TWorkShifts",
          "id": 6,
          "fields": {
            "FOwnsObjects": true,
            "FListHelper": []
          }
        },
        "Fovertimes": {
          "type": "WorkCenterShift.TWorkShifts",
          "id": 7,
          "fields": {
            "FOwnsObjects": true,
            "FListHelper": []
          }
        },
        "Fmax_week_end_overtime": 0,
        "Fmax_weekly_overtime": 0,
        "Fmax_daily_overtime": 0,
        "FProcess_id": -1,
        "FNext_process_id": -1,
        "FID": 4,
        "FCreatedDate": 45737.9440097106,
        "FUserID": 0,
        "FRefCount": 0
      }
    },
    "FItemUnit": "KG",
    "FFGCode": "INV001",
    "FInstruction": "",
    "FRawMaterials": {
      "type": "ProcessRawMaterial.TProcessRawMaterials",
      "id": 8,
      "fields": {
        "FOwnsObjects": true,
        "FListHelper": [
          {
            "type": "ProcessRawMaterial.TProcessRawMaterial",
            "id": 9,
            "fields": {
              "FCode": "Inv2",
              "FQuantity": 1,
```

```
                        "FItemUnit": "PCS",
                        "FItemDesc": "",
                        "Fbatch_number": "",
                        "Fcritical": false,
                        "FID": 1,
                        "FCreatedDate": 45737.9440097106,
                        "FUserID": 0,
                        "FRefCount": 0
                    }
                }
            ]
        }
    },
    "FNextProcess": {
        "type": "Process.TNextProcess",
        "id": 10,
        "fields": {
            "FQuantity": 0,
            "FProcess": null,
            "FID": -1,
            "FCreatedDate": 45737.9440097106,
            "FUserID": 0,
            "FRefCount": 0
        }
    },
    "FBSCode": "",
    "FResiduCode": "",
    "FCritical": false,
    "FProcess_times": {
        "type": "Process.TProcessTimes",
        "id": 11,
        "fields": {
            "FOwnsObjects": true,
            "FListHelper": []
        }
    },
    "FCriticalQuantity": 0,
    "FworkCenterId": -1,
    "FnextProcessId": -1,
    "FID": 1,
    "FCreatedDate": 45737.9440097106,
    "FUserID": 0,
    "FRefCount": 0
}
```

```
},
{
    "type": "Process.TProcess",
    "id": 12,
    "fields": {
        "FRoute": {
            "type": "Route.TRoute",
            "ref": 1
        },
        "FOutputQuantity": 20,
        "FWorkCenter": {
            "type": "WorkCenter.TWorkCenter",
            "id": 13,
            "fields": {
                "FWorkCenterName": "",
                "FMachines": {
                    "type": "Machine.TMachines",
                    "id": 14,
                    "fields": {
                        "FOwnsObjects": true,
                        "FListHelper": []
                    }
                },
                "FNumberOfShift": 0,
                "Fwork_shift": {
                    "type": "WorkCenterShift.TWorkShifts",
                    "id": 15,
                    "fields": {
                        "FOwnsObjects": true,
                        "FListHelper": []
                    }
                },
                "Fovertimes": {
                    "type": "WorkCenterShift.TWorkShifts",
                    "id": 16,
                    "fields": {
                        "FOwnsObjects": true,
                        "FListHelper": []
                    }
                },
                "Fmax_week_end_overtime": 0,
                "Fmax_weekly_overtime": 0,
                "Fmax_daily_overtime": 0,
                "FProcess_id": -1,
```

```
        "FNext_process_id": -1,
        "FID": 5,
        "FCreatedDate": 45737.9440097106,
        "FUserID": 0,
        "FRefCount": 0
      }
    },
    "FItemUnit": "KG",
    "FFGCode": "INV001",
    "FInstruction": "",
    "FRawMaterials": {
      "type": "ProcessRawMaterial.TProcessRawMaterials",
      "id": 17,
      "fields": {
        "FOwnsObjects": true,
        "FListHelper": [
          {
            "type": "ProcessRawMaterial.TProcessRawMaterial",
            "id": 18,
            "fields": {
              "FCode": "FIN1010",
              "FQuantity": 2,
              "FItemUnit": "PCS",
              "FItemDesc": "",
              "Fbatch_number": "",
              "Fcritical": false,
              "FID": 2,
              "FCreatedDate": 45737.9440097106,
              "FUserID": 0,
              "FRefCount": 0
            }
          },
          {
            "type": "ProcessRawMaterial.TProcessRawMaterial",
            "id": 19,
            "fields": {
              "FCode": "RW1009",
              "FQuantity": 3,
              "FItemUnit": "PCS",
              "FItemDesc": "",
              "Fbatch_number": "",
              "Fcritical": true,
              "FID": 3,
              "FCreatedDate": 45737.9440097106,
```

```
                        "FUserID": 0,
                        "FRefCount": 0
                    }
                }
            ]
        }
    },
    "FNextProcess": {
        "type": "Process.TNextProcess",
        "id": 20,
        "fields": {
            "FQuantity": 10,
            "FProcess": {
                "type": "Process.TProcess",
                "ref": 3
            },
            "FID": -1,
            "FCreatedDate": 45737.9440097106,
            "FUserID": 0,
            "FRefCount": 0
        }
    },
    "FBSCode": "",
    "FResiduCode": "",
    "FCritical": false,
    "FProcess_times": {
        "type": "Process.TProcessTimes",
        "id": 21,
        "fields": {
            "FOwnsObjects": true,
            "FListHelper": []
        }
    },
    "FCriticalQuantity": 0,
    "FworkCenterId": -1,
    "FnextProcessId": -1,
    "FID": 2,
    "FCreatedDate": 45737.9440097106,
    "FUserID": 0,
    "FRefCount": 0
}
}
]
}
```

```
        },
        "FInstruction": "",
        "FBSCode": "",
        "FResiduCode": "",
        "FSuspended": false,
        "FFGQty": 0,
        "FID": 99,
        "FCreatedDate": 45737.9440096759,
        "FUserID": 0,
        "FRefCount": 0
    }
}
```


Info:

-source:

-minimalist examples, unused information has been deleted.

- Return:string

function delete_route

- Purpose:
- Parameters:
  - Route_id:integer
  - test_tx:TjbTx=nil
- Return:string

function load_outstanding_so_det

- Purpose:
- Parameters:
  - test_tx:TjbTx=nil
- Return:string (separated by **^**)
  - SOID
  - SONO
  - SODate
  - Estimated Ship Date
  - Customer Name
  - ItemNo
  - Item Description
  - Item Unit
  - Seq

    - Quantity
    - Qty Shipped
    - On Production
    - On hand
    - On Sales
    - MPQ
    - Unit Ratio
    - Transfer No (only if using MPO)
    - QtyMPO (only if using MPO)
    - OutstandingSOQty (only if using MPO)
    - TransferID (only if using MPO)
    - Seq (only if using MPO)

function load_bom_by_finishing_good

- Purpose:
- Parameters:
    - Fg_code:string
- Return:string

function load_raw_material_list

- Purpose:
- Parameters:
    - Route_id:integer
- Return:string

function load_RMOut_batch_number_list

- Purpose:
- Parameters:
    - Item_no:string
- Return:string

function Load_RM_IA_Value_batch_number_list

- Purpose:
- Parameters:
    - item_no: string
- Return: string

function Load_RM_IA_Qty_batch_number_list

- Purpose:

- Parameters:
    - item_no: string
- Return: string

function get_scheduling_wo_sql

- Purpose:
- Parameters:
    - deadline_time:TDateTime
    - Work_center_id:integer
- Return:string

function load_scheduling_work_orders

- Purpose:
- Parameters:
    - deadline_time:TDateTime
    - Work_center_id:integer
- Return:string

function load_process_times

- Purpose:
- Parameters:
    - Process_id:integer
- Return:string

function load_scheduling_machines

- Purpose:
- Parameters:
    - Work_center_id:integer
    - test_tx:TjbTx=nil
- Return:string

function get_valid_user_id

- Purpose: To Get UserID
- Parameters:
    - User_name:string
    - User_password:string
- Return:integer

function get_product_structure_output_quantity_by_process

- Purpose:
- Parameters:
    - Process_id:integer
- Return:currency

function get_critical_quantity

- Purpose:
- Parameters:
    - Process_id:integer
- Return : Currency

function get_process_needed_input_qty

- Purpose:
- Parameters:
    - Process_id:integer
    - test_tx:TjbTx=nil
- Return:currency

function save_machine

- Purpose:
- Parameters:
    - machine_data:TJSONValue
    - test_tx:TjbTx=nil
- Return:string

function load_machine

- Purpose:
- Parameters:
    - machine_id:Integer
- Return:TJSONValue

procedure load_machine_from_db

- Purpose:
- Parameters:

    - var machine:TMachine
    - Machine_id:integer
    - loadWorkShift:boolean=True
- Return: none

function save_work_center

- Purpose:
- Parameters:
    - work_center_data:TJSONValue
    - test_tx:TjbTx=nil
- Return:string

function load_work_center

- Purpose:
- Parameters:
    - Work_center_id:integer
- Return:TJSONValue

procedure load_work_center_from_db

- Purpose:
- Parameters:
    - var work_center:TWorkCenter
    - Work_center_id:integer
    - loadWorkShift:boolean=True
- Return : none

function load_unscheduled_wo

- Purpose:
- Parameters:
    - test_tx:TjbTx=nil
    - FilterFrom:TDatetime=0
    - FilterTo:TDateTime=0
- Return:TJSONValue

function load_item_info

- Purpose:to get information about description, unit, ratio, minimumQty, onSales, onOrder, onHand and onProduction based on selected ItemNo.
- Parameters:
    - const item_no : string

- Return:TJSONValue

function save_requisition

- Purpose:
- Parameters:
    - requisition_data:TJSONValue
    - test_tx:TjbTx=nil
- Return:string

function load_schedule

- Purpose:
- Parameters:
    - Schedule_id:integer
- Return:TJSONValue

function load_MFG_user_profiles

- Purpose:
- Parameters:
    - User_id:integer
    - var user_name:string
    - var user_password:string
    - var user_level:integer
- Return:string

function save_mfg_user_profiles

- Purpose:
- Parameters:
    - User_id:integer
    - User_profiles:string
    - new_password:string
- Return:string

function load_user_name_list

- Purpose:
- Parameters:
- Return :string

function load_inventory_item_list

- Purpose: Load RM when Add button is clicked.
- Parameters:
- Return:string

function get_production_result_id

- Purpose:
- Parameters:
- Return:Integer

function get_options_value

- Purpose:
- Parameters:
    - SParam:String
    - DefValue:String
- Return:String

procedure set_options_value

- Purpose:
- Parameters:
    - SParam:String
    - SValue:String
- Return:none

procedure load_work_center_machines

- Purpose:
- Parameters:
    - Wc_id:integer
    - var machines:TMachines
    - loadWorkShift:boolean=True
- Return:None

function loadRequestRM

- Purpose:
- Parameters:
    - scheduleId:integer
- Return:string, with order: (ScheduleID, Production_ID, ID WOID, RMNO, ITEMUNIT, QUANTITY, Taken, OnHand, i.ItemDescription)

function get_field_title

- Purpose:
- Parameters:
    - usedSql:TjbSql
- Return:TJSONArray

function get_record_data

- Purpose:
- Parameters:
    - usedSql:TjbSql
- Return: TJSONArray

function sql_data_to_json

- Purpose:
- Parameters:
    - Sql_text:string
    - dbIdx:integer=-1
- Return:TJSONObject

function sql_commit_data

- Purpose:
- Parameters:
    - Sql_to_execute_list:string
    - dbIdx:integer=-1
- Return:TJSONObject

Scheduling

function loadCCRList

- Purpose:
- Parameters:
- Return:TJSONValue

function loadCCRDrums

- Purpose:
- Parameters:
    - deadline:TDateTime
    - Ccr_id:integer

- Return:TJSONValue

function loadCCRMachines

- Purpose:
- Parameters:
    - Ccr_id:integer
    - loadWorkShift:boolean=True
- Return:TJSONValue

function loadCCRProductStructures

- Purpose:
- Parameters:
    - deadline:TDateTime
    - Ccr_id:integer
- Return:TJSONValue

function loadSubordinateProductStructures

- Purpose:
- Parameters:
    - deadline:TDateTime
    - Ccr_id:integer
- Return:TJSONValue

function loadSubordinateWorkCenters

- Purpose:
- Parameters:
    - deadline:TDateTime
    - Ccr_id:integer
    - loadWorkShift:boolean=True
- Return:TJSONValue

function loadProductStructure

- Purpose:
- Parameters:
    - productStructureId:integer
- Return:TJSONValue

function loadWorkCenter

- Purpose:
- Parameters:

    - workCenterId:integer
    - loadWorkShift:boolean=True
- Return:TJSONValue

function saveSchedulerResult

- Purpose:
- Parameters:
    - productionScheduleData:TJSONArray
- Return:TJSONObject

function getMPOStatus

- Purpose:
- Parameters:
- Return:Boolean

function GetScheduleWOStatus

- Purpose:
- Parameters:
    - ScheduleID: integer
- Return: Boolean

TFinaCloudMethod

function calculateDiscount

1. Purpose:
2. Parameters:
    a. itemDiscPC : string
    b. brutoUnitPrice : Currency
    c. decimalDigit : Integer
3. Return: TJSONObject

function calculateTax

1. Purpose:

2. Parameters:
    a. taxData : string
    b. inclusiveTax : Boolean
    c. tax1Code:String
    d. tax2Code : string
    e. cashDiscount:Currency
    f. tax1Rate:Currency
    g. tax2Rate : Currency
    h. databaseIndex : Integer
3. Return : TJSONObject

function getProductInfo

1. Purpose:
2. Parameters: none
3. Return :TJSONObject

function fetchJSONTransaction

1. Purpose:
2. Parameters:
    a. id : Integer
    b. Datamodule:string
    c. transType : string
    d. databaseIndex : Integer
3. Return: TJSONObject

function modifyJSONTransaction

4. Purpose: To insert or update data to datamodule in FINA
5. Parameters:
    a. modifiedJSON : string
6. Return: TJSONObject
    a. If success: {success:'Edit Done'}
    b. If failed: {error: [Error Message]}

Format data

```
{"_parameters":["{
      \"id\"          : number,
```

```
      \"dataModule\"  : string,
      \"transType\"   : string,
      \"dbIndex\"     : number,
      \"data\":{
            \"header\": \"append\" | \"edit\",
            \"master\":{
                  \"field1\":xxx,
                  \"field2\":yyy,
                  ...,
            },
            \"detail\":[
                  {
                        \"field_a\":xxx,
                        \"field_b\":yyy,
                        ...
                  },
                  ...
            ]
      }
}"]}
```

Explanation

12. Because the json data is inside the parameters value, so every double quote must be escaped (\").
13. dbIndex is a parameter to determine which database to be modified.
14. header:
    a. **append** for inserting new record
    b. **edit** for editing existing records.
15. For all fields, refer to each table's field name.
16. List of available dataModules and their transType:

Transaction Name	DataModule Name	transType
Journal Voucher	TdmJV	GLJV
Other Payment	TdmOtherPayment	GLOP
Other Deposit	TdmOtherDeposit	GLOD
Sales Order	TdmSalesOrder	N/A
Delivery Order	TdmARInvoice	ARDLV
Sales Invoice	TdmARInvoice	ARINV

Sales Invoice from DO	TdmARInvoice	ARSTM
Customer Receipt	TdmARPayment	ARPMT
Sales Return	TdmCreditMemo	N/A
Purchase Order	TdmPurchaseOrder	N/A
Receive Item	TdmAPInvoice	APRCV
Purchase Invoice	TdmAPInvoice	APINV
Vendor Payment	TdmAPCheque	APCHQ
Item Transfer	TdmStockTrans	N/A
Item Request	TdmItemRequest	N/A
Job costing	TdmJobCost	PRJOB
Inventory Adjustment	TdmAdjustItem	ITADJ

Example

Json data for Sales Invoice


```
{"_parameters":["{
	\"publicIp\":\"localhost\",
	\"id\":\"78\",
	\"dataModule\":\"TdmARInvoice\",
	\"transType\":\"ARINV\",
	\"dbIndex\":\"2\",
	\"data\":{
		\"header\":\"edit\",
		\"master\":{
			\"GUID\":\"{00686F52-7DBB-48FA-A702-C65203E04D39}\",
			\"ARINVOICEID\":\"78\",
			\"INVOICENO\":\"scy5\",
			\"CUSTOMERID\":\"8\",
			\"WAREHOUSEID\":\"0\",
			\"GETFROMDO\":\"0\",
			\"GETFROMSO\":\"0\",
			\"INVOICEDATE\":\"2020-06-11\"
			,\"RATE\":\"1\",
			\"INVOICEAMOUNT\":\"27300\",
			\"PAIDAMOUNT\":\"0\",
			\"TERMDISCOUNT\":\"0\",
```

\"RETURNAMOUNT\":\"0\",
\"OWING\":\"27300\",
\"TERMSID\":\"1\",
\"GLPERIOD\":\"6\",
\"GLYEAR\":\"2020\",
\"PRINTED\":\"0\",
\"TAX1ID\":\"3\",
\"TAX1CODE\":\"B\",
\"TAX1RATE\":\"30\",
\"CHEQUEDATE\":\"2020-06-11\"
,\"SHIPTO1\":\"Budi Sungkono\",
\"SHIPTO2\":\"Jl. Budi Kemuliaan no:456\",
\"SHIPDATE\":\"2020-06-11\",
\"GLHISTID\":\"116\",
\"PAYMENT\":\"0\",
\"CASHDISCOUNT\":\"0\",
\"CASHDISCPC\":\"0\",
\"TEMPLATEID\":\"20\",
\"ARACCOUNT\":\"110301\",
\"GETFROMOTHER\":\"0\",
\"DELIVERYORDER\":\"0\",
\"FISCALRATE\":\"1\",
\"TAX1AMOUNT\":6300,
\"TAX2AMOUNT\":0,
\"INCLUSIVETAX\":\"0\",
\"CUSTOMERISTAXABLE\":\"1\",
\"TAXFORMNUMBER\":\"20.0000scy5\",
\"FREIGHT\":\"0\",
\"RECONCILED\":\"0\",
\"INVFROMSR\":\"0\",
\"TAXDATE\":\"2020-06-11\",
\"REPORTEDTAX1\":\"0\",
\"REPORTEDTAX2\":\"0\",
\"ROUNDEDTAX1AMOUNT\":\"6300\",
\"ROUNDEDTAX2AMOUNT\":\"0\"
,\"PRINTEDTIMES\":\"0\",
\"BillTo1\":\"Budi Sungkono\",
\"BillTo2\":\"Jl. Budi Kemuliaan no: 456\",
\"BillTo4\":\"Jakarta \",
\"BillTo5\":\"\",
\"EXTENDEDID\":\"577\",
\"EXCHANGED\":\"0\",
\"ExtendedID\":\"577\",
\"SUBTOTAL\":\"21000\",
\"TOTALINVOICE\":27300},
\"detail\":[
{
\"ARINVOICEID\":\"78\",
\"SEQ\":\"1\",

```
\"ITEMNO\":\"10001\",
\"ITEMOVDESC\":\"Busi\",
\"ITEMUNIT\":\"PCS\",
\"QUANTITY\":\"2\",
\"BRUTOUNITPRICE\":\"5500\",
\"UNITPRICE\":\"5500\",
\"TAXCODES\":\"B\",
\"ITEMHISTID\":\"168\",
\"ITEMDISCPC\":\"0\",
\"Amount\":\"11000\",
\"UNITRATIO\":\"1\",
\"TAXABLEAMOUNT1\":\"11000\",
\"TAXABLEAMOUNT2\":\"0\",
\"ITEMCOST\":\"5500\",
\"ITEMCOSTBASE\":\"5500\",
\"WAREHOUSEID\":\"0\",
\"InvoiceNo\":\"scy5\",
\"WarehouseName\":\"PSB\",
\"SNDummyField\":\"<empty>\",
\"EXTENDEDID\":\"578\",
\"IS_PROMO\":\"0\",
\"GUID\":\"{163EE113-FC9C-4BC8-8942-FD5AABC2BA3C}\",
\"ExtendedID\":\"578\",
\"DEPTID\":\"\",
\"PROJECTID\":\"\"
},
{
\"ARINVOICEID\":\"78\",
\"SEQ\":\"2\",
\"ITEMNO\":\"10001\",
\"ITEMOVDESC\":\"Busi\",
\"ITEMUNIT\":\"BOX\",
\"QUANTITY\":\"1\",
\"BRUTOUNITPRICE\":\"10000\",
\"UNITPRICE\":\"10000\",
\"TAXCODES\":\"B\",
\"ITEMHISTID\":\"172\",
\"ITEMDISCPC\":\"0\",
\"Amount\":\"10000\",
\"UNITRATIO\":\"10\",
\"TAXABLEAMOUNT1\":\"10000\",
\"TAXABLEAMOUNT2\":\"0\",
\"ITEMCOST\":\"10000\",
\"ITEMCOSTBASE\":\"10000\",
\"WAREHOUSEID\":\"0\",
\"InvoiceNo\":\"scy5\",
\"WarehouseName\":\"PSB\",
\"SNDummyField\":\"<empty>\",
\"EXTENDEDID\":\"654\",
```

\"IS_PROMO\":\"0\",
\"GUID\":\"{72385BFC-92A5-40F7-A6B0-A6206CFD153E}\",
\"ExtendedID\":\"654\"
}
]
}
}"]}

function deleteJSONTransaction

1. Purpose:
2. Parameters:
    a. modifiedJSON : string
3. Return: TJSONObject

TOrmMethod

POST lookup

- Purpose: To return an array of {value, label} for select component in the form.
- Parameters:
    - {
      index: number,
      table: string, //table lookup name
      key: string, //primary key field name
      display: string //lookup field name
      }
- Return: [{value:1, label:'xx'}, {value:2, label:'yy'}, ….]

POST orm

- Purpose: To handle all method for Object-Relational Mapping
- Parameters:
    - {
      dbIndex: integer,
      Method: ['get' | 'put' | 'delete'],
      object:{
      [table name]:{
      primarykey:xxx,
      field1:'',
      field2:'',

```
                    ….
                },
                parameters:{}
            },

        }
```

- index = to determine which database to use
- Method = see below explanations.
- object:
    - **Table name** will be used as the table name
    - value is an object that:
        - Key = field name
        - Value = field value
- parameters: used only form **list** method.

Method **get**

- Purpose: to get an object json from database
- How to use:
    - The first key of the table object will be used as the primary key.
    - The rest of the keys will be used as field names.
- Example:

{index:1, method:"get", object:{employees:{id:3, name:", salary:0, email:"}} will be converted into:

Select id, name, salary, email from employees where id=3

To get list of detail object in Master-Detail ORM, use this json format:

```
{
      Index: integer,
      Method: 'getd',
      object:{
            [table name]:{
                  ID:xxx,
                  field1:'',
                  field2:'',
                  ….
                  [detail_table_name]:[
                        {
                              ID:yyy,
                              [table_name]_ID:xxx,
                              d_field1:'',
                              d_field2:'',
                              …
                        }
                  ]

            },
            detail:'[detail_table_name]',
            parameters:{}
      }
}
```

The Result will be:
Select id, d_field1, d_field2 from detail_table_name where table_name_ID=xxx

To get a list of detail objects in Master-Detail with non-ORM-compliance:

1. Define the detail table name and the master key field in the detail object.
2. Make sure the masterkeyid in detail array object **is not** the first key
3.

This is the example:

```
{
      Index: integer,
      Method: 'getd',
      object:{
            [table name]:{
                  [masterkeyid]:xxx,
                  field1:'',
                  [detail_table_name]:[
                        {
                              seq:yyy,
                              [masterkeyid]:xxx,
                              d_field1:'',
                        }
                  ]

            },
            detail:{
                tableName: '[detail_table_name]',
                masterKey: '[masterkeyid]'
            },
            parameters:{}
      }
}
```

The Result will be:
Select seq, d_field1 from detail_table_name where **masterkeyid**=xxx

Method **put**

- Purpose: to insert/update an object json into database
- How to use:
    - Value of [**tablename]** can be an object or an array
    - If the value is an object, it will generate SQL like this: **Update or Insert into TableName (key1, key2, …) values (value1, value2, …)**
    - If the value is an array, the above sql will be generated as many as the array element.
    - **Key1** should be the primary key of the table
    - put can also be used for master-detail tables. When edit, detail will be deleted first before appended.

Method **delete**

- Purpose: To delete a/some record(s) of a table.
- How to use:
    - object: {[tablename]:{key:value, ….}}
    - [TableName] will be used as the table to be deleted.
    - The key(s) of [tablename] will be used as the where clause to delete.
- Example:
    - {index:0, method:'delete', object:{users:{id:3}} will be converted into: **delete from users where id=3**.
    - {index:0, method:'delete', object:{users:{id:3, status:'1'}} will be converted into: **delete from users where id=3 and status='1'**.

Method **deletemd**

- Purpose: To delete a master-detail table
- How to use:
    - Object: {[MasterTableName]:{key:value, [DetailTableName]:[]}
    - [MasterTableName] is the master table name to be deleted
    - [DetailTableName] is the detail table name to be deleted

POST **setTemplate**

- Purpose: To insert or update to Template Table
- Parameters:
    - dbindex:string
    - filename:string
    - template: string

POST list

- Purpose: To query data from a table
- Parameters:

{

length: number, //determine maximum records to get. Default: unlimited
start: number, //determine how many batches to skip. 0= get from first.
order: [
{
column: number, //index of column to be sort
dir: [‘asc’ | ‘desc’], //asc = a-z, desc = z-a
}
],
filter: [
junction: [‘’ | ‘and’ | ‘or’], //First filter should be ’’, the rest either **and** / **or**.
column: string, //column to be filtered. Follow the format of **columns|data**.
operator: [‘=’, ‘<>’, ‘>’, ‘<’, ‘>=’, ‘<=’, ‘between’, ‘like’, ‘in’],
value1: string|number, //for first filter value, can use select clause
value2: string|number, //for **between** operator
filter:[] //can be indented
],
columns:[
data: string, //see explanation below
title: string //will be shown in Table Header.
]

}

- Column selection:
  - For simple select: [table name].[field name]
  - For join to another table:
    `TBL_NAME.ID_FK.LOOKUP_ID.LOOKUP_TBL.LOOKUP_FIELD`
  - For joining to another foreign key
    `TBL_NAME.ID_FK~LOOKUP_ID.LOOKUP_TBL.LOOKUP_FIELD.LOOKUP2_ID.LOOKUP_TBL3.LOOKUP3`
  - First table name will be used as the primary table.
  - First join table’s alias will be A, second will be B, and so on.
    - For example: [{column:’tblX.b’}, {column:’a.cid.id.tblY.d’}, {column: ‘a.eid.id.tblZ.f’}]
    - Will be transformed to:

```
Select tblX.b, A.d, B.f
From tblX
Join tblY A on A.id=tblX.cid
Join tblZ B on B.id=tblX.eid
```

- Filter
  - column:
    - Closure with bracket () if the where clause is an sql select statement. For example: column: (select count(*) from a where a.id=tbl1.id)

POST block

- Purpose: To execute sql block
- Parameters: Object of:

```
{
  dbIndex:number,
  method:'block',
  object:[execute_block_string]
}
```

Example:
http://IP:212/fina/rest/TOrmMethod/%22block%22
Body:

```
{
  dbIndex:1,
  method:'block',
  Object:'execute block returns (a) as begin a=1; suspend; end'
}
```

GET img

- Purpose: to retrieve image from table
- Parameters:
  - dbIndex → which database index to query from
  - Table → table name
  - KeyField → Key field to be in where clause
  - imageField → Blob field containing the image to be sent
  - Value → key Value. If the value is string, it must be contained in %27 (single quote).
  - formatField (optional) → image format field. If empty, the image is assumed to be a jpeg image.

Example:
To query logo field from item table in FINA, use this url. Remember to put the query parameter **json=false**.

http://IP:212/fina/rest/TOrmMethod/img/1/item/itemno/logo/%27BW001%27/format_logo?json=false

POST img

- Purpose: to update an image to a table
- Parameters:

```
{
  dbIndex:string,
  img : string,
  tableName:string
  imgCol:string,
  keyCol:string,
  keyValue:string,
  formatCol:string,
  formatValue:string
}
```

Explanation:

1. **dbIndex** referring to which database index in FINA API.
2. **img** is the image data to be saved in base64 format.
3. **tableName** is the name of the table to be updated.
4. **imgCol** is the image column to be updated
5. **keyCol** is the primary key of the table to be updated.
6. **keyValue** is the key value of the record to be updated.
7. **formatCol** is the optional key to update the image file format. If it is not provided, FINA API will assume the image format is jpeg.
8. **formatValue** is the optional value that must exist if formatCol exists. The value will determine the image file format.

GET genId

- Purpose: to generate ID from a firebird generator
- Parameters:
    - generator_Name: string
    - dbIndex: string

Security

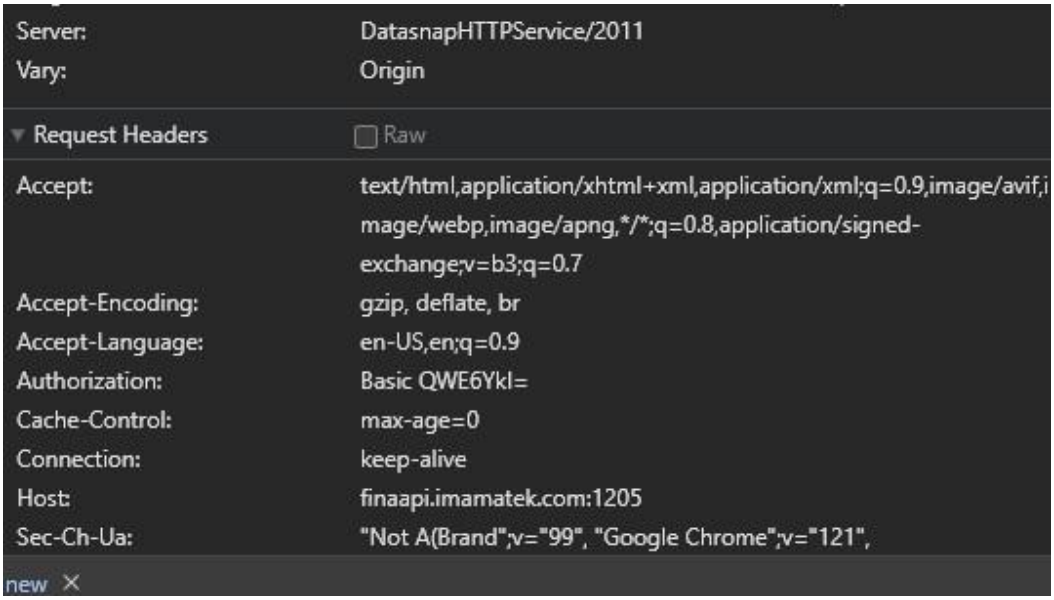
Basic Authentication

1. Isikan nilai pada serverSetting.ini :

   [Main]
   AuthUser=Aa
   AuthPassword=bB

   Kosongkan atau hapus kedua key ini bila tidak mau pakai basicAuth.

2. Lakukan Request ke FINA API dengan mengirimkan username dan password sebagai attribute ‘Authorization’ pada header request.
   a. Format nya -> Basic <base64encode of username:password>
   b. Perhatikan ada tanda titik dua (:) antara username dan password
   c. Base64encode untuk Aa:bB adalah QWE6Ykl=



   d.
   e. Bila tidak ditaruh sebagai attributes dari header, maka browser akan meminta hal ini ketika request pertama kali dilakukan ke FINA API, untuk selanjutnya akan di simpan di cache browser tersebut.

Cross-Origin Resource Sharing (CORS) & Preflight request

1. Isikan nilai pada serverSetting.ini, contoh :

[Main]
AllowOrigin=https://biz-a.id, https://www.imamatek.com:1205

Delimiter = comma (,)
Kosongkan atau hapus key ini bila tidak mau pakai CORS.

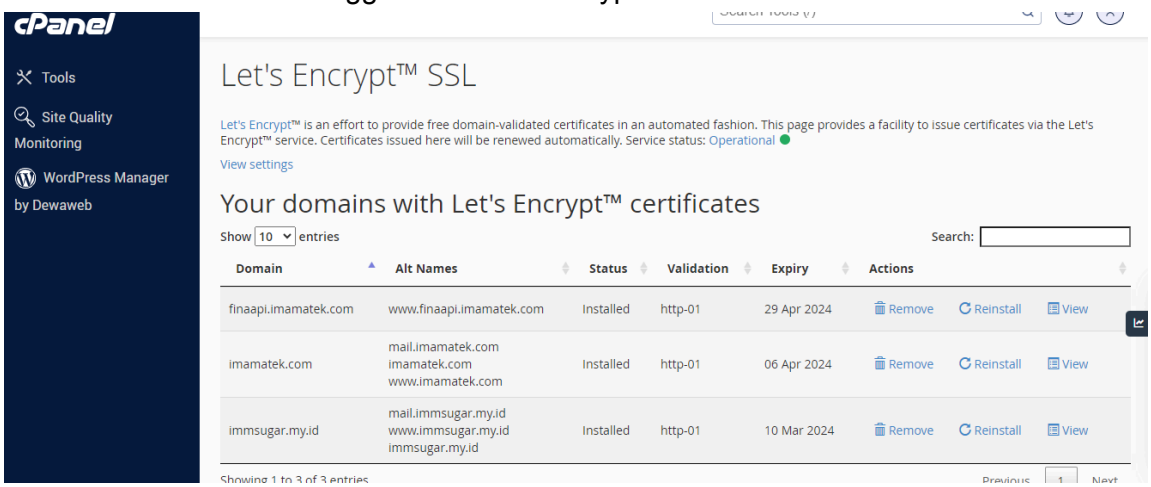
2. CORS Origin sudah bisa multiple Origin, dengan delimiter ",". Bisa di set di serverSetting.ini untuk FINA API, section [Main], key AllowOrigin=a.com,b.com,...'
3. FINA API hanya akan melayani request dengan attribute ‘Origin:https://biz-a.id’ pada Header Request.
4. Saat ini sudah bisa multiple origin yang diperbolehkan, mohon perhatikan port yang digunakan, bila ada ya.
5. Pastikan selalu lower case yang dipakai
6. Perhatikan juga protocol yang dipakai, baik itu http ataupun https
7. Custom Headers yang diperbolehkan adalah ‘biza-ce’ dan ‘content-type’
8. Methods yang diperbolehkan adalah semua (‘*’)
9. Preflight Check Request cache adalah 24 jam (bergantung pada agent \ browser yang dipakai)
10. Response header 'Access-Control-Allow-Method' di set dengan nilai '*'
    a. Semua method kita perbolehkan, baik itu 'post', 'put', 'head' dan sebagainya
11. Response header 'Access-Control-Allow-Headers' di set dengan nilai 'api-content-encoding', 'api-accept-encoding', 'content-type'
    a. ini agar requester (BizA Client, MFG, dan sebagainya), bisa menambahkan custom header ke request yang akan mereka kirimkan ke FINA API.
12. Response header 'Access-Control-Expose-Headers' di set dengan nilai 'api-content-encoding', 'content-type'
    a. ini diperlukan, agar Agent (browser) memperbolehkan header tersebut untuk diakses oleh aplikasi seperti Angular (BizA Client).
13. Response header 'Access-Control-Max-Age' di set dengan nilai '86400'
    a. CORS Max Age saat ini di set 24 jam, agar sisi agent bisa melakukan preflight request cache bila memungkinkan.
    b. Namun hal ini tergantung masing - masing agent (chrome max 2 jam, firefox max 24 jam).
    c. Nilai nya selama ini selalu nilai default, yaitu 5 detik.
    d. Dengan ini maka CORS di FINA API sudah bisa handle preflight check dengan custom headers, dan mengurangi jumlah preflight request yang duplikat.

SSL \ TLS

1. Pastikan memiliki dedicated IP / Public IP, contoh : 36.93.175.197
2. Pastikan memiliki FQDN / subDomain, contoh : finaapi.imamatek.com.
3. Arahkan FDQN tersebut ke Public IP di point 1, contoh : buat sebuah DNS Record dengan type A

Type	Name	Content	Proxy status	TTL	Actions
A	finaapi	36.93.175.197	DNS only	Auto	Edit
A	www.finaapi	36.93.175.197	DNS only	Auto	Edit

4. Buat SSL dengan trusted CA untuk FQDN tersebut untuk mendapatkan cert file, key file dan root file. Contoh : menggunakan let's encrypt dari cPanel



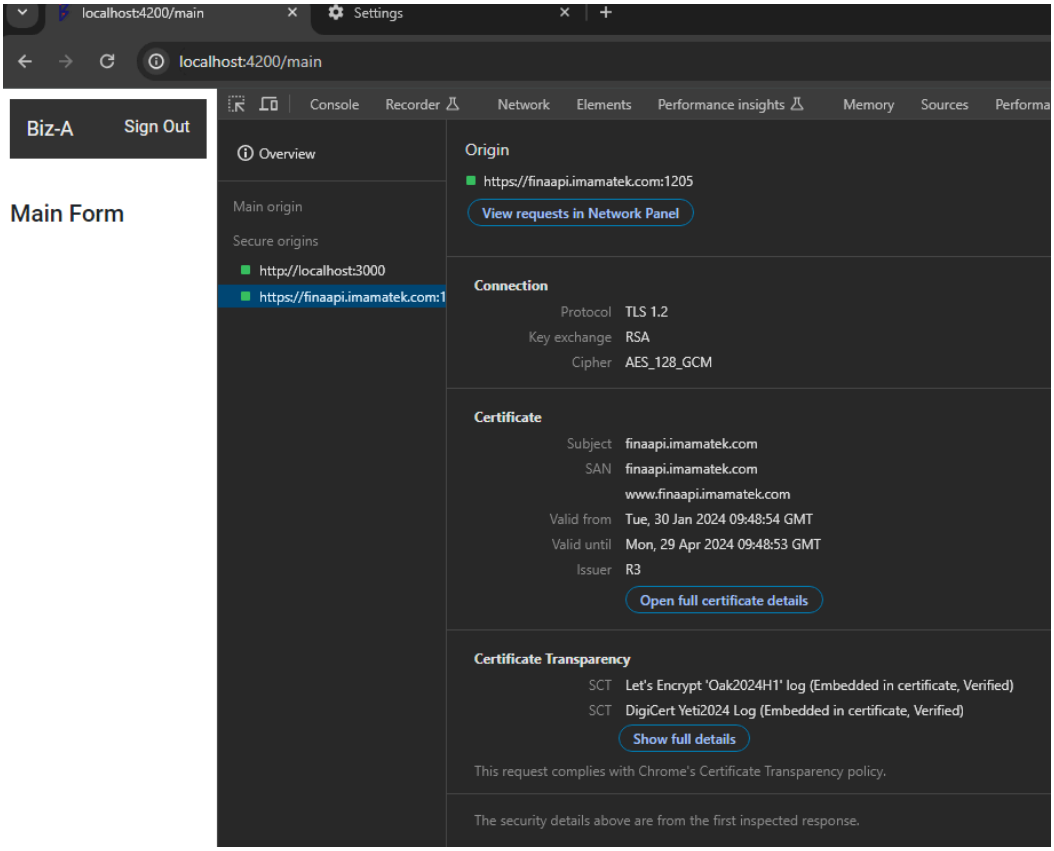
5. Isikan nilai pada serverSetting.ini :

   [Main]
   httpPort=1205
   CertFile=finaapi.imamatek SSL\cert.pem
   RootCertFile=finaapi.imamatek SSL\root.pem
   PrivateKeyFile=finaapi.imamatek SSL\key.pem
   TLSVersion=1.2

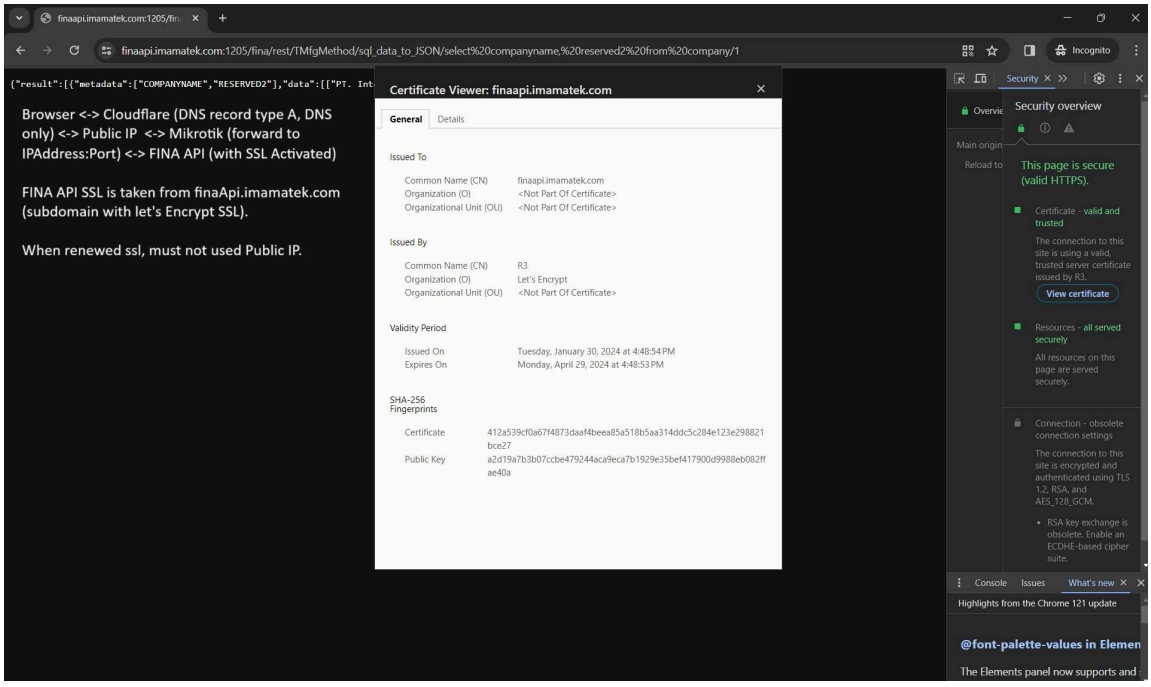
   Key TLSVersion disesuaikan dengan format SSL yang dipakai di point 4.
   Key ini bisa bernilai 1, 1.1, 1.2 (untuk saat ini TLS 1.3 belum di support). Default = 1.2
   Kosongkan atau hapus key CertFile, RootCertfile, PrivateKeyFile bila tidak mau pakai SSL

   Note :
   ====
   a. Format yang dipakai untuk SSL file adalah "**base-64 encoded x.509**"
   b. Bila ada perubahan di isi *.pem files, pastikan FINA API di buka ulang.
   c. Bila akan di pakai oleh aplikasi Biz-A (saat ini Biz-A menggunakan Axios untuk request ke FINA API). Axios memerlukan RootCertFile wajib dalam **full chain certificate**. Isi dari rootCertfile dimulai dari rootCert, intermediateCert, Cert.
      Contoh :
      -----BEGIN CERTIFICATE-----

MIIFazCCA1OgAwIBAgIRAIIQz7DSQONZRGPgu2OCiwAwDQYJKoZIhvcNAQELBQAw
… // ini adalah root Certificate
emyPxgcYxn/eR44/KJ4EBs+lVDR3veyJm+kXQ99b21/+jh5Xos1AnX5iItreGCc=
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
MIIFFjCCAv6gAwIBAgIRAJErCErPDBinU/bWLiWnX1owDQYJKoZIhvcNAQELBQAw
…// ini adalah intermediate Certificate
nLRbwHOoq7hHwg==
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
MIIFDzCCA/egAwIBAgISBBf8LeUljRu7d1OW1hyB/7DjMA0GCSqGSIb3DQEBCwUA
…// ini certificate untuk FQDN finaapi.imamatek.com
MIuW
-----END CERTIFICATE-----



6. Lakukan Request ke FINA API dengan FDQN dan port yang dipakai, contoh : https://finaapi.imamatek.com:1205/fina/rest/TMfgMethod/sql_data_to_JSON/select companyname, reserved2 from company/1



Performance

I. Compress

1. Dilakukan apabila size data yang akan dikembalikan lebih besar dari 1 KB (KiloBytes) dan request header 'api-accept-encoding'
2. Bila request header 'api-accept-encoding' mengandung nilai 'deflate', maka lakukan data compression dengan ZLib compression level clMax (9)
3. Bila request header 'api-accept-encoding' mengandung nilai 'deflatejson', maka lakukan JSON compression dengan point III.
4. Size data yang di transfer (compressed) dan resources size (uncompressed) harusnya lebih kecil bila compress dilakukan. Namun perlu perhatikan ada kondisi di mana bisa terjadi sebalik-nya, hal ini karena pada sebuah response ada terdapat data - data lain, selain data yang dikembalikan, seperti response header yang lain
5. Harus selalu mengirimkan raw byte agar agar size kecil, bisa diterima oleh metode apapun pada sisi Agent (browser), tidak perlu handle special character maupun encode dan decode. Bila menggunakan base64 size malah membesar menjadi 33%.
6. Sudah disiapkan juga agar FINA API bisa melakukan compression Ketika response yang dikembalikan adalah octet, namun sepertinya belum ada metode di FINA API yang melakukan hal ini.

II. Decompress

1. Bila request header 'api-content-encoding' mengandung nilai 'deflate', maka lakukan data decompression dengan ZLib compression level clMax (9)
2. Bila request header 'api-content-encoding' mengandung nilai 'deflatejson', maka lakukan JSON decompression dengan point 5

III. JSON Compression (JSON Object ke Array of Records)

1. FieldNames di simpan ke key '_m'
2. Setiap record berupa array of data
3. Semua record akan di simpan ke key '_d' berupa array of array
4. Format JSON untuk array of records adalah

```
{
    "_m" : [fieldName1, fieldName2, ...],
    "_d" : [[recordData], [recordData], [recordData], ...]
 }
```

5. semua RecordData wajib memiliki jumlah data yang sama dan urutan yang sesuai dengan fieldNames yang ada di keys "_m"
6. Bila object yang tidak memiliki keys yang diperlukan, maka nilai dari field tersebut akan diisikan dengan !e ("there does not exists" symbol)
7. Deep compression, dimana akan diterapkan di semua key dari JSON Object, termasuk bila object di dalam object (all nested object)

IV. JSON Decompression (Array of Records ke JSON Object)

1. untuk setiap Array of Records
2. Ubah setiap record yang ada di key '_d' (array of array) menjadi 1 buah object
3. keyName di ambil dari key '_m'
4. Format JSON untuk array of records adalah

```
{[
  {fieldName1 : recordData1[fieldName1], fieldName2:
recordData1[fieldName2]},
  {fieldName1 : recordData2[fieldName1], fieldName2:
recordData2[fieldName2]},
  ....
]}
```

5. Semua RecordData akan memiliki jumlah key yang sama dan urutan yang sesuai dengan fieldNames yang ada di keys "_m"
6. Bila recordData memiliki nilai '!e' maka field tersebut tidak akan ditambahkan sebagai key untuk object yang sedang di proses. Perhatikan '!e' berbeda dengan Null.

7. Deep decompression, dimana akan diterapkan di semua array of Records dari JSON Object, termasuk bila array of records di dalam array of records (all nested Array of Records)

History

Date	Description	PIC
15 Sep 2023	Added json2fina for TdmJCRO Added orm.deletemd Changes: orm.put will delete detail before append.	**MAS**
16 Sep 2023	Added TUserApiDataset	**MAS**
21 Oct 2023	POST list: 1. filter property can have optional filter 2. Return null as null object, not 'null' string	**MAS**
13 Dec 2023	Add template	**MAS**
5 Jan 2024	POST list: filter can use select clause	**MAS**
10 Jan 2024	POST calcDiscount is added	**MAS**
1 Feb 2024	FINA API Security v1.0.3.733	**AA**
17 Jul 2024	POST Template : ● Optional parameter "userId" and "imageQuality" ● Multiple page / image	**AA**
2 Sep 2024	CORS : ● Access-Control-Allow-Headers ● Access-Control-Allow-Methods ● Access-Control-Max-Age	**AA**
27 Feb 2025	Penambahan fitur performance Perbaikan CORS dan penambahan pre-flight.	**AA**